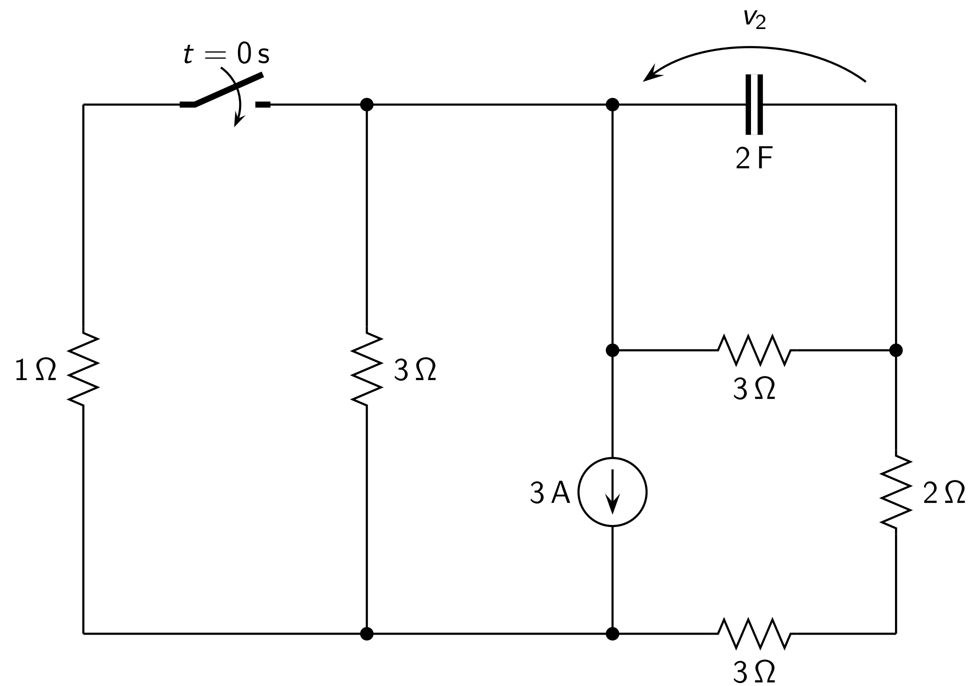
Problem: find the expression of $v_2(t)$ for $t > 0$.



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Solution

- Time constant: $\tau = 3.943\,\text{s}$;
- Initial conditions before switching: $v_2(0^-) = \frac{-27}{11}\,\text{V}$;
- Initial conditions after switching: $v_2(0^+) = \frac{-27}{11}\,\text{V}$;
- Steady-state solution: $v_2(\infty) = \frac{-27}{35}\,\text{V}$;
- Solution for $t > 0$:

$$v_2(t) = \left(-1.683\, e^{-t/\tau} + \frac{-27}{35}\right)\,\text{V}$$

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